Chapter 11 Notes

Section 0

Notes

Note that elementarily equivalence only requires pure formulas to be equivalent for the following reason: $A_1 \neq A_2$ may be true, and if it is, then if $a \in A_1$ and $a \notin A_2$, then there are formulas either true or false in (A_1, R_1) (formulas involving a as a constant) where truth in (A_2, R_2) isn't even defined.

Exercise 0.1

Suppose π is an isomoprhism from (A_1, R_1) to (A_2, R_2). We show by complete induction on the degrees of formulas that for any pure formula $\phi(x_1, ..., x_n)$ and any elements $a_1, ..., a_n$ of A_1, the sentence $\phi(a_1, ..., a_n)$ is true in (A_1, R_1) iff $\phi(\pi(a_1), ..., \pi(a_n))$ is true in (A_2, R_2), and, for any pure sentence ϕ, ϕ is true in (A_1, R_1) iff ϕ is true in (A_2, R_2).

In the base case, Pab (for elements a, b in A_1) is true in (A_1, R_1) iff aR_1b iff $\pi(a)R_2\pi(b)$ (because π is an isomorphism) iff $P\pi(a)\pi(b)$ is true in (A_2, R_2). Thus the base case holds.

Now suppose that the result holds for all formulas of degree less than n. That is, for any pure formula $\phi(x_1, ..., x_m)$ (of a degree less than n) and any elements $a_1, ..., a_m$ of A_1, $\phi(a_1, ..., a_n)$ is true in (A_1, R_1) iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2), and for any pure sentence ϕ, ϕ is true in (A_1, R_1) iff ϕ is true in (A_2, R_2). On the one hand, we consider a pure formula with free variables of degree n. That is, we consider some pure formula $\phi(x_1, ..., x_m)$ and elements $a_1, ..., a_m$ of A_1. There are then multiple cases.

In the first case, $\phi(x_1, ..., x_m) = (\psi \wedge \gamma)$ for some pure formulas ψ and γ of degrees less than n. Either (1) neither ψ nor γ are sentences; (2) ψ is a sentence and γ is not; (3) ψ is not a sentence and γ is a sentence. We consider the first case - the other two are nearly identical. In that case, because neither ψ nor γ are sentences, $\psi = \psi(y_1, ..., y_k)$ for some $k < n$. Similarly $\gamma = \gamma(z_1, ..., z_l)$ for some $l < n$. Then $\phi(x_1, ..., x_m) = (\psi(y_1, ..., y_k) \wedge \gamma(z_1, ..., z_l))$. Then $\phi(a_1, ..., a_m) = (\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ where each $b_i = a_j$ for some j (and each $c_i = a_j$ for some j). Then $\phi(a_1, ..., a_m)$ is true in (A_1, R_1) iff $(\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ is true in (A_1, R_1) iff $(\psi(\pi(b_1, ..., \pi(b_k))) \wedge \gamma(\pi(c_1), ..., \pi(c_l)))$ is true in (A_2, R_2) (by the induction assumption) iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2).

The second case, where $\phi(x_1, ..., x_m) = \sim\psi(x_1, ..., x_m)$ (for some formula $\psi(x_1, ..., x_m)$, is similar to the first case.

The third case is when $\phi(x_1, ..., x_m) = (\exists x)\phi(x, x_1, ..., x_m)$. Then $(\exists x)\phi(x, x_1, ..., x_m)$ is true iff $\phi(a, a_1, ..., a_m)$ is true for some $a \in A_1$, iff $\phi(\pi(a), \pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2) iff $(\exists x)\phi(x, \pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2).

On the other hand, we consider a pure sentence ϕ of degree n. The cases here are nearly identical to the above three cases - we give the case for existential quantification.

In that case, $\phi = (\exists x)\psi(x)$ for some ψ. Then ϕ is true in (A_1, R_1) iff $\psi(a)$ is true in (A_1, R_1) for some $a \in A_1$ iff $\psi(\pi(a))$ is true in (A_2, R_2) (by the induction assumption) iff $(\exists x)\psi(x)$ is true in (A_2, R_2).

Section 1

Theorem 1.1

In the proof, it is remarked that there are only c formulas with constants in S, hence S' has no higher cardinality than S. In the non-denumerable case, this does not at all seem obvious to me, so here is an explanation.

Let F_n be the set of all formulas (with constants in S) of length n. Informally, the set of all sequences of length n (where $n > 0$) made up of only $(,), \exists, \sim, \vee$, and constants in S, has the same cardinality as C^n, where $C = \cup\{\{(\}, \{)\}\}, \{\sim\}, \{\vee\}, \{\exists\}, \}$ (and C has cardinality c). Now, $\overline{\overline{C \times C}} = c$, as $C \times C \cong C$ (by the fundemental theorem of cardinal arithmetic). Obviously the set of all formulas (with constants in S) of length n is not equal to C^n, but it is (informally) a subset of it. So $F_n \preceq C^n \cong c$. But it is obvious that there is a subset of F_n of cardinality c, meaning $c \preceq F_n$, and thus $F_n \cong c$ (by the Bernstein-Schröder Theorem).

Thus we know that, for each i, $\overline{\overline{F_i}} = c$. Now, the set of all formulas with constants in S is equal to $F : \cup\{F_1, F_2, ..., F_n, ...\}$. We can show that $F \cong c$ in the following way. We show that $F \cong \mathbb{N} \times F_j$ (for any j; which one doesn't matter), and because $\mathbb{N} \times F_i \cong c$ (by the law of multiplicative absorpition), this would imply $F \cong c$. Well, we know that each F_i is well ordered (because they have some cardinality at all). Note that F is a union of pairwise disjoint sets. So, for each $x \in F$, let $\phi(x) = \langle i, y\rangle$ where i is the F_i such that $x \in F_i$, and, if x is the αth element of F_i, let y be the αth element of F_j. Because ϕ is a 1-1 correspondence between F and $\mathbb{N} \times F_j$, it follows that $F \cong c$, which is what we set out to show.

The above argument is certainly informal, and not technically correct (for instance, we are using cardinality theorems on things that are not even formally sets). If we wanted to be more formal about it, we could try to represent the connectives (and parentheses) as sets. To do this, consider any set S. We must be able to find some five sets a_0, a_1, a_2, a_3, a_4 outside of S (or else V would be a set). Then we let $(= a_0,) = a_1, \exists = a_2, \sim = a_3$, and $\vee = a_4$, and then the set of all formulas with constants in S would just be a subset of $H \times H$ where $H = \cup\{\{(\}, \{)\}\}, \{\exists\}, \{\sim\}, \{\vee\}, S\}$ (which has cardinality c), and we can proceed in a similar way as we did in the informal argument.

Also, we are using the fundemental theorem of cardinal arithmetic (FTCA), which requires the axiom of choice. Actually, we can alter the proof of the FTCA (in chapter 9) so that $A \cong A \times A$ follows if A is infinite and we only know that that A itself has a choice function. This works as, importantly, Kuratowski's maximal principle (which is used in the proof of Theorem 6.1 of Chapter 9) does not require AC to hold; only that A is closed under chain unions and has a choice function. Throughout the proof of the FTCA (and the preceeding lemmas), whenever we use the fact that AC holds, for whatever set we need to have a choice function, we can show that it is also well ordered.

The above argument can also be expanded so that variables are accounted for; i.e, findining a denumerable number of elements not in A (this can be done, by the axiom of substitution).

In particular, the additional assumptions to get the result in the preceeding paragraph would be (the below theorems and lemmas are from chapter 9):
Theorem 6.1: A is well orderable and A' is well orderable.
Theorem 6.2: A and B are well orderable (actually we don't need to know that B is well orderable).
Lemma 6.3: M and M' are well orderable.
Theorem 6.4: A is well orderable.

Exercise 1.1

This is just a matter of redifining truth over A so that a sentence X with constants in A is true over A iff it is true in the relational system (A, R). Then the rest of the proof is the same.

Exercise 1.2

Assuming the axiom of choice, this means that every infinite set has a denumerable subset. Now, recall that the Skolem-Löwenheim theorem states that if ϕ (having no constants) is satisfiable in any infinite relational system at all, then

it is satisfiable in a denumerable relational system. Well, suppose ϕ is satisfiable in some infinite relational system (A, R). Then A can be well ordered (as the axiom of choice is assumed). Suppose that ϕ has n free variables, meaning it can be written as $\phi(x_1, ..., x_n)$. Because it is satisfiable in (A, R), $\phi(a_1, ..., a_n)$ is true in (A, R) for some $a_1, ..., a_n$ in A. Then, by the axiom of choice, there is a denumerable subset A_0 of A which includes all of $a_1, ..., a_n$. By Exercise 1.1, there is some denumerable subset B of A such that B reflects A, and B is an extension of A_0. This means that, because $\phi(a_1, ..., a_n)$ is true in (A, R), $\phi(a_1, ..., a_n)$ will be true in (B, R). Thus ϕ is satisfiable in the denumerable domain B.

Section 2

Exercise 2.1

This exercise does not seem solvable (whether or not A is denumerable). For instance, suppose that no elements of A are $\in$-connected; that is, for all elements x and y of A, $x \notin y$ and $y \notin x$ both hold. Then, for all elements a and b of A, Pab is always false (over A). An obvious induction then shows that, for any formula $\phi(x)$ with constants in A, $\phi(a)$ has the same truth value for all elements a of A. What that means is that the only definable sets over A are $\emptyset$ and A itself.

However, if A was extensional, the situation would be different. Then (as can easily be seen), $\phi(x)$:

$$\forall y(y \in x \equiv y \in a)$$

for some element a of A, defines the subset $\{a\}$ of A. Thus, because A is denumerable, it must at least have a denumerable number of definable subsets. But the number of definable subsets of A is at most denumerable (Reason: There are only a denumerable number of formulas with constants in A, hence there are at most a denumerable number of formulas that define some subset of A, hence there are at most a denumerable number of definable subsets of A). Thus, by the Bernstein-Schröder theorem, $\overline{\overline{\mathcal{F}(A)}} = \aleph_0$.

I would assume that the exercise was really meant to assume that A was extensional.

Exercise 2.2

Suppose that A is finite and extensional and n is the number of elements of A. We will show that all finite subsets of A are definable (this argument also carries over to Exercise 2.1); that is, we show that all elements of $\mathcal{P}(A)$ are definable, and because $\mathcal{P}(A)$ has 2^n elements, this means that $\overline{\overline{\mathcal{F}(A)}} = 2^n$.

For any elements a and b of A, let $a = b$ be an abbreviation for the formula

$$(1)\ \forall x(x \in b \equiv x \in a)$$

It should be noted why $=$ can only be defined because A is extensional. That is, (1) is true iff $a = b$. Because A is extensional, for all a, b in A, if $a \neq b$, then $a^* \neq b^*$. But recall that a^* is the set of all $x \in A$ such that $x \in a$ (and similarly for b^*), as we are considering truth *over* A, that is, in the relational system $(A, \in)$. Obviously, if $a = b$, then (1) is true (over A). On the other hand, if (1) is true over A, then it must be that $a^* = b^*$, and thus, by the assumption of extensionality, $a = b$. The important thing is, if A was not assumed to be extensional, (1) could be true for all a and b in A (whether or not $a = b$), because it could be that a^* and b^* are both the emptyset!

Now consider any finite subset S of A. Either it is empty or it isn't.

If it is empty, then it can be defined by the formula $x \in a \wedge x \notin a$ (for some $a \in A$).

If it isn't empty, then it can be enumerated in some sequence $a_1, ..., a_k$ of elements of A. Then the formula

$$x = a_1 \vee ... \vee x = a_k$$

defines S.

Thus $\overline{\overline{\mathcal{F}(A)}} = 2^n$.

Exercise 2.3

We suppose that every non-empty subset of A which is B-definable over A contains at least one element of B. We prove by complete induction that B reflects A (the proof closely follows the proof of Lemma 1.2).

Proof. Call a formula (with constants in B) *good* if B reflects A with respect to it.

(1) It is obvious that any atomic formula is good.

(2) It is obvious that the negation of any good formula is good.

(3) It is obvious that the conjunction of any two good formulas is good.

(4) Now consider a formula of the form $(\exists x)\phi(x, y_1, ..., y_n)$ $x, y_1, ..., y_n)$ are the free variables of ϕ) such that $\phi(x, y_1, ..., y_n)$ is good. We must show that the formula $(\exists x)\phi(x, y_1..., y_n)$ is good, i.e, that for any elements $b_1, ..., b_n$ of B, the sentence $(\exists x)\phi(x, b_1, ..., b_n)$ is true over B iff it is true over A.

(a) Suppose it is true over B. Then it is obviously also true over A.

(b) Conversely, suppose the sentence $(\exists x)\phi(x, b_1, ..., b_n)$ is true over A. Now, $\phi(x, b_1, ..., b_n)$ defines some B-definable subset of A (because it has only constants in B). But, by the hypothesis, this means that there is some $b \in B$ such that $\phi(b, b_1, ..., b_n)$ is true over A. But, by the induction assumption, this means $\phi(b, b_1, ..., b_n)$ is also true over B. Thus $(\exists x)\phi(x, b_1, ..., b_n)$ is true over B. □

Exercise 2.4

Suppose that B is an elementary subsystem of A. Let C be any subset of A that is B-definable over A (by some $\phi(x)$). We are to show that $C \cap B$ is definable over B. In fact, $C \cap B$ is definable by $\phi(x)$ over B. For, because $\phi(x)$ B-defines C, all constants of $\phi(x)$ are in B. But then, because B is an elementary subsytem of A, this means that for all $b \in C \cap B$, $\phi(b)$ is true over A iff $\phi(b)$ is true over B. But $\phi(b)$ is true over A for all $b \in C \cap B$. And it cannot be true for any element outside of $C \cap B$ (For, if $\phi(a)$ were true over B for some $a \notin C \cap B$ (and a must be in B, so $a \notin C$), then, because B is an elementary subsytem of A, $\phi(a)$ would be true over A, where $a \notin C$, contradicting the fact that $\phi(x)$ defines C). Thus $\phi(x)$ defines $C \cap B$ over B.

Section 3

Notes

It is remarked (in the addendum to part 1) that a well ordering W of A could be definable, in particular, by some formula $\phi(x, y)$. But the book has yet to provide a definition of what it means for a set to be defined by a formula with *two* free variables. I would assume it means: $\phi(x, y)$ defines (over K) the set of all ordered pairs $\langle x, y\rangle$ such that $\phi(x, y)$ is true over K. This would match up with how it is used in the book.

Theorem 3.3

The proof offered in the book of Theoerem 3.3 is not correct. In particular, consider the proof of Fact 1. In it, we are trying to show that a_0 is B-describable over A. Now, the formula claimed to prove this is

$$\phi(x) \wedge (\forall y)(\phi(y) \supset w(x, y))$$

Now, in order for this to show that a_0 is B-describable over A, that formula must only include constants in B (by definition). However, we have no guarantee that $w(x, y)$ has only constants in B, so this formula is incorrect.

I suspect that this proof can be salvaged - particularly, if a well ordering W (of A) is definable over A at all, then it seems likely that it must be defined by some formula $\phi(x, y)$ with *no* constants. But I am not able to prove this (or find a counterexample).

Exercise 3.1

Lemma 0.1. For any set A, any $\phi(x, y)$ (with constants in A), and any sets x, y in A, if $x^* = y^*$, $x \neq y$, and $\phi(x, y)$ is false (over A), then

(a) We suppose that the relation $x = y$ is definable over A, and are to show that for any subset B of A, $B \subseteq B^*$. First, note that A must be extensional. For suppose it weren't. Then, for some x, y in A such that $x \neq y$, $x^* = y^*$. Now, suppose $\phi(x, y)$ defines the relation $x = y$ over A. Then we would

(b) Suppose A is extensional. Then (as we have seen in the exercises of section 2) $x = y$ is definable over A. Hence, by part (a) of this exercise, $B \subseteq B^*$.

(c) Note that the proof of Theorem 3.3 in the book is not correct (see the section "Theorem 3.3" of these notes). So, for this exercise, let us assume the theorem can in fact be proven. So suppose some well ordering of A is definable over A and the relation $x = y$ is definable over A. Thus, by part (a) of this exercise, $B \subseteq B^*$ for any subset B of A. Consider any infinite subset A_0 of A. By Theorem 3.3, A_0^* reflects A. Also $A_0 \subseteq A_0^*$, hence $A_0 \preceq A_0^*$. But also $A_0^* \preceq A_0$ (Reason: $x \in B^*$ iff $\{x\}$ is definable over A by some formula $\phi(x)$ with constants in B. But, because A_0 is infinite, there are $\overline{\overline{A_0}}$ formulas with constants in A_0, meaning there are at most $\overline{\overline{A_0}}$ elements of A_0^*). Thus, by the Bernstein Schröder Theorem, $A_0 \cong A_0^*$, meaning A_0^* is a superset of A_0 of no higher cardinality than A_0 such that A_0^* reflects A.